Photon

Planned integration

Connect owner approval and technician communications to the same service records the dashboard uses. Start with a message round trip and one approved notification call.

Current handoff and target flow

The deployed backend creates one queued technician-contact action with each approved work order. The production owner can record contact and acceptance manually. The local demo simulates both steps. A Photon sender, receiver, and calling worker still need implementation and provider verification.

Step	Planned behavior
Owner message	Send a concise observation, uncertainty, proposed inspection and unique request reference.
Owner reply	Map a verified sender and conversation to a pending request and version. Call the same backend decision operation as the dashboard.
Approved technician contact	Claim the committed contact action, check the work order and recipient, then place an outbound call with a short spoken summary.
Technician response	Record explicit acceptance or decline tied to the work order. Record the original repair report and verified sender before updating work progress.
Owner updates	Reflect the same work-order and verification milestones in messages and in the dashboard.

Example message format

Use unambiguous replies such as APPROVE REQ-104, DECLINE REQ-104, ACCEPT WO-104 or REPORT WO-104 followed by findings. These references illustrate the proposed message format. The backend currently uses its request and work-order IDs.

Provider setup and processing

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Prove the two provider paths

01 Messaging: connect a test project and line. Send to an authorized test recipient, receive the reply through the chosen verified ingress, and record the provider message ID.

02 Voice: use a Spectrum project-owned iMessage line and a compatible SIP application. Use the provider's TLS trunk settings, verify caller ID, and confirm audible playback with the authorized test recipient.

Photon Spectrum outbound calling

Webhooks, scope and durable state

Use verified HTTP webhooks or a hosted stream consumer. Native Spectrum webhooks verify an HMAC over the original raw request body. Preserve raw bytes at ingress and deduplicate on provider message ID before applying decisions. Confirm callback and job lifetimes on the selected host.

Resolve the owner or technician using configured identities, then verify building, request version, work order, and allowed state. Ask for clarification on ambiguous replies. Replayed, expired, superseded, unknown or mismatched requests must leave the approval and dispatch state intact.

Store provider event IDs, recipient, attempt timestamps, statuses, and errors. Reuse stable send IDs where supported. Reconcile an uncertain call attempt before retrying. Keep secrets server-side and use a separate test project or simulator for rehearsal.

Photon Spectrum webhooks

Photon retry and recovery guidance

Acceptance and fallback

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A connected call and an accepted work order are separate

The call identifies the building, approved work order, observed issue, requested inspection, and how to respond. A connected call, voicemail, busy line or timeout records a call outcome. Only an explicit technician response can record acceptance. A repair report still awaits verification.

Test	Expected outcome
Approve from the owner dashboard or verified reply	The same backend gate yields one work order and one committed contact action.
Decline, duplicate reply or stale request	Decline authorizes no call. Repeated or invalid events cannot create extra work or sends.
Provider fails before or after acknowledgment	Preserve the outcome, retry only when safe, and reconcile uncertain delivery.
Technician accepts and reports findings	The existing work order advances. Owner sees the findings and can verify the repair.
Inbound or outbound provider unavailable	Run the existing dashboard workflow with manually recorded milestones, or use labeled simulation.

The first live MVP is complete after a verified owner message round trip, one approved outbound call with audible playback, an explicit technician acceptance, and a repair report reflected in the dashboard.